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ENGINEERING
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INGENIEURSWESE

M&M Skripsie Topics

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1 Mechanical and/or Mechatronic Topics at MM

1.1 Design and Mechatronics Division

Design, build and test of a PV support structure vibration exciter

Reference	: JB1
Supervisor	: Mr Johann Bredell
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

In the structural design of utility-scale PV support structures, it is important to know the modal frequency and damping properties. These characteristics are used in wind studies to determine wind load coefficients and evaluate aeroelastic stability. It is required to design a portable dynamic exciter that can be attached to the structure and dynamically excite the structure at a range of frequencies. The measured response can be used to determine dynamic properties. The exciter must be powered from a 12 V power supply, like a battery or a car. A control interface must be provided to select a single frequency or perform a frequency sweep. (Mechanical or Mechatronic)

Key modules (Scientific and Engineering Knowledge):

Electrical Drive Systems (324) | Machine Design A (314) | Vibration and Noise (354)

Design of a re-deployable tensile structure for solar PV

Reference	: JB3
Supervisor	: Mr Johann Bredell
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

The design of a re-deployable, semi-permanent fixed-tilt structure for supporting PV modules is required. Current practice relies on steel frames for such structures. Alternative concepts based on tensile structural principles, including steel wire rope and/or inflatable systems, must be investigated. In addition, methods of non-permanent anchoring are to be evaluated. Prototype structures are to be fabricated to demonstrate functionality. (Mechanical)

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Machine Design B (344) | Strength of Materials W (334)

Software tool for slew drive sizing in PV single-axis trackers

Reference	: JB4
Supervisor	: Mr Johann Bredell
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Single-axis tracker (SAT) structures are widely used in utility-scale solar PV plants to improve energy yield relative to fixed-tilt systems. However, SATs are susceptible to aeroelastic instabilities and must move to a safe stow position to prevent divergent dynamic response once the wind speed exceeds a certain threshold. EPC companies require a software tool for the preliminary sizing of the high-cost slew drives and DC motors. The tool must account for both normal tracking operation over the design life and emergency stow scenarios, where the tracker must stow while exposed to wind loads. A slew drive may actuate a single tracker row or two rows via a mechanical linkage. An industry partner will provide representative input parameters. In addition to the report, a concise user manual which explains assumptions and calculation procedures must be written. This project involves developing and demonstrating such a tool.

Key modules (Scientific and Engineering Knowledge):

Electrical Drive Systems (324) | Machine Design A (314) | Modelling (334)

Inert Gas Chamber Development for High-Temperature Materials Testing

Reference	: LC2
Supervisor	: Mr Llewellyn Cupido
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: Finite Element Methods (FEM)

Description:

This project involves designing and analysing an inert-gas-controlled-atmosphere chamber for high-temperature materials testing. The chamber will enable micro-hardness testing and other materials characterisation at elevated temperatures (up to 1000°C) while protecting specimens from oxidation through an argon atmosphere. Building on analytical and numerical foundations established in 2025, this continuation project addresses the complete engineering challenge: thermal management, atmospheric control, structural integrity, and device integration.

Core Engineering Activities: - Thermal-Structural Analysis: Validate and extend computational models. Investigate thermal fields, structural deflections, and innovative thermal management approaches, including magnetic field effects on heat transfer. - Atmospheric Control Design: Design complete inert gas handling systems, including supply, purging, flow control, oxygen monitoring, and safety interlocks.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Heat Transfer A (414) | Machine Design A (314)

Advanced Multi-Sensor Precision Temperature Control System

Reference	: LC3
Supervisor	: Mr Llewellyn Cupido
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Develop an advanced precision temperature control system that integrates multi-sensor state estimation with coordinated multi-zone heating to achieve improved thermal stability and spatial uniformity for high-temperature material testing.

Key Technical Areas

1. **Multi-Sensor State Estimation** Design an intelligent sensing system providing spatial temperature awareness:
 - Sensor placement optimisation
 - Sensor fusion algorithms
 - Spatial reconstruction
 - Uncertainty quantification
 - Fault detection & isolation

2. **Multi-Zone Temperature Control** Implement coordinated control of independent heating zones:
 - Zone architecture
 - Control strategy
 - Thermal coupling
 - Multi-objective optimisation
 - Real-time implementation

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electronics (245)

Design of a Declination Drive Sub-System for a Satellite Ground Station (2 students: 1 mechanical, 1 mechatronic))

Reference	: NM1
Supervisor	: Prof Nawaz Mahomed
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

A satellite ground station has been successfully built and operated. The SGS uses an AC motor with VSD to position the antenna in the azimuth plane. The control system consists of a Rasp Pi with already-developed Python code for converting satellite position data (TLE) to observer coordinates in terms of azimuth and declination. A GUI running on the Pi can be used to select the satellite to be tracked. The ground station uses a parabolic dish antenna designed to operate at 1.7 GHz. Satellites communicating in this band have already been successfully tracked and packets downloaded (and decoded). Currently, it can only track in the azimuth plane.

The ground station can be deployed in TLE (two-line-element) or signal strength mode.

The aim now is to add a declination axis (subsystem) to the existing ground station. The system will likely comprise a servomotor powered from the existing control box. The software needs to be developed further to allow for the declination positioning. The project will involve development of Python code, design of the declination motor structure, the design of any electronic components for interfacing the declination drive to the Rasp Pi.

This project requires the detailed design and software development for the declination drive. The build part depends on available time beyond the scope of the project.

In addition, the project will investigate the difference in position of the satellite based on the TLE data (which uses a propagator model) and signal strength (which is assumed to be the actual position). This will give a "residual" of the position vector, which will allow further understanding of the perturbations experienced by a satellite (such as atmospheric drag, solar radiation and gravitational fields).

The project entails some learning in the fields of orbital mechanics and RF engineering. The latter will involve calibration tests to be conducted on the antenna to establish radio frequency (RF) parameters, using a spectrum generator and spectrum analyser.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Control Systems (354) | Machine Design A (314)

Meteoritic Fragmentation using Light Curve Analysis (1 student)

Reference	: NM2
Supervisor	: Prof Nawaz Mahomed
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

From optical observation images of meteors, the light intensity / luminosity can be measured as a function of time, giving what is called a light curve. A light curve provides important information regarding the fragmentation / breakup and ablation (mass loss due to viscous forces) of a meteoroid. FLARE (Fireball Luminosity & Ablation Reconstruction Engine) was developed in 2025 to convert meteor captured image stacks into light curves, and to identify the type of fragmentation undergone by a meteor. This project aims to further improve the prediction accuracy of FLARE, which will be made available for public use.

The project aims to generate light curves for a number of meteors / fireballs captured through the allsky camera network in the Western Cape, as well as cameras linked to the Global Meteor Network. The light curve analysis (LCA) will be aimed at identifying the type of fragmentation and ablation (based on well-established classifications) experienced by different meteoroids. Using such data, the project will attempt to quantify the size of a meteoroid, based on images from multiple cameras.

Key modules (Scientific and Engineering Knowledge):

Data Analytics (Eng) (344) | Experimental Techniques (414)

Fragmentation of Heterogeneous Granular Composite Materials under Thermal Shock (2 students)

Reference	: NM3
Supervisor	: Prof Nawaz Mahomed
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Meteoroids entering the earth's atmosphere experience breakup / fragmentation at very low stagnation pressures. This phenomenon cannot be explained, and scientists currently only use scaling laws or empirical models to model fragmentation during atmospheric flight.

To arrive at a scientifically sound basis for this phenomenon, it is proposed that meteoroids are modelled as composite granular materials, a collection of discrete particles that may be a mixture of different materials or a combination of a granular solid within a matrix. Such materials behave as quasi-brittle materials, and the stress/strain relationship can then be modelled using Coulomb-Mohr Theory. Since atmospheric entry at hyper-velocity (Mach 20 upwards) results in a rapid change in temperature due to viscous drag and radiation, the thermal stress induced needs to be included in the stress/strain model. This project will attempt to model the thermal stress / strain field due to thermal shock, which leads to (rapid) crack propagation in the granular material, hence leading to breakup / fragmentation.

This project is divided into two distinct components. The first involves the experimental reconstruction of artificial meteoritic materials, and using such samples to undertake compression tests coupled to Digital Image Correlation (DIC), to analyse the volumetric strain fields and crack growth. This requires insight into stress/strain relationships and fracture mechanics. This work also includes the use of 3D scanning data on actual meteorite samples.

The second part deals with the modelling and simulation of the stress-strain fields using suitable computational simulation tools. This requires the development of a representative volume element (RVE) using data from the first part of the project.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Strength of Materials W (334) | Thermofluid Dynamics (344)

3D object ranging and orientation technology (2 students: 1 mechanical, 1 mechatronic)

Reference	: NM4
Supervisor	: Prof Nawaz Mahomed
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Light detection and ranging methods have rapidly developed over the past decade to precisely measure distances and create 3D images of objects. However, these technologies, such as those used in autonomous vehicles, require high power and onboard processing capacity, and are therefore not suitable for certain applications.

This project aims to develop a low-power technology capable of creating a 3D spectral signature of an object, which can be used to determine the range and orientation of an identified object. The idea is to use the solar radiation reflectivity of metallic materials (such as aluminium) to identify the target body, and to further use the intensity/magnitude/brightness of reflected light to determine orientation. The latter will generate a "light curve" (magnitude vs orientation in 3D) as a spectral signature of the object.

The project will require the design and build of a two degree-of-freedom (rotation along two axes) system that will support and move the object into different orientations with respect to the observer. A light source, emulating that of solar radiation, will be directed at the object. A photodetector at the observer will initially scan the environment and then lock onto the target based on the target material's spectral properties. The photodetector will then measure the magnitude of the reflected light to generate a 3D map of the object's orientation, i.e. the 3D spectral image.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Modelling and Design of a Robotic End-effector to Perform Complex Maintenance Activities in the Manufacturing Industry.

Reference	: BR1
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Maintenance plays a crucial role for transport manufacturing industry. Most maintenance activities are still performed manually, thus resulting in safety concerns and inconsistencies due to human errors. Thus, more intelligent, and automated type of maintenance solutions are required to guarantee safety of operations as well as continuous availability of equipment during operations. This project aims to model and simulate a robotic end-effector to perform automated maintenance operations in the transport manufacturing industry. The Computer Aided Design (CAD) will be carried out using CAD design software's under different loading conditions. Furthermore, kinematics models of the designed end-effector will be modelled in MATLAB environment, including the kinematic motion and trajectory of the robotic end-effector along the X-Y and Z coordinates.

Key modules (Scientific and Engineering Knowledge):

Machine Design B (344) | Mechatronics (424) | Modelling (334)

Design of a Reconfigurable Electric Press Brake Machine for Customised Sheet Metal Bending Operations

Reference	: BR2
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Sheet metal manufacturing is a fundamental part of modern industry, transforming raw metal into essential components for a wide range of products. Traditional press brake machines often lack the flexibility to adapt to varied and complex bending tasks required in modern manufacturing industries. To address this limitation, this project aims to design a reconfigurable electric press brake machine tailored for customized sheet metal bending operations. The proposed machine will incorporate reconfigurable manufacturing systems (RMS) principles to enable operators to adjust its configuration based on specific bending requirements. This design will use electric actuation system to enhance energy efficiency, precision, and environmental sustainability as compared to hydraulic counterparts. This will include control algorithms are integrated to ensure consistent bending angles, reduced spring back effects, and high repeatability. In addition, this design should be able to achieve quick tooling changes and process optimization, making it suitable for low-volume, high-mix production environments. CAD software will be used to model the machine, MATLAB Simulink will be used to design and simulate the control system. A simulation model of the proposed solution will also be developed to demonstrate its capabilities in bending different sheet metal geometries.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design B (344) | Mechatronics (424)

Design of a Modular Open Architecture Controller for a Reconfigurable Machine Tool

Reference	: BR3
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Controllers for dedicated machines, while effective for specific tasks, often face several limitations that hinder their adaptability and performance in modern manufacturing environments. These controllers are not suitable for reconfigurable manufacturing systems (RMS). RMS demands highly adaptable and modular controllers capable of integrating new hardware and software modules without significant reprogramming or downtime. This project aims to design a modular open-architecture controller (MOAC) for a reconfigurable machine tool (RMT) to enable seamless plug-and-play functionality. The proposed MOAC will integrate modular design and open communication protocols to ensure compatibility with various machine tool configurations and components. The controller will be designed with a layered architecture comprising hardware abstraction, communication, and application layers, ensuring scalability and ease of integration. Plug-and-play functionality will be achieved through standardized interfaces, automatic detection of modules, and dynamic reconfiguration capabilities. The controller will be validated through simulations as well as implementation on the existing RMT prototype machine.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (214) | Machine Design B (344) | Mechatronics (424)

Design of a Reconfigurable Sorting and Inspection Machine in Automotive Manufacturing Industries

Reference	: BR5
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

In automotive manufacturing industries, sorting and inspection systems play a critical role in ensuring product quality and consistency. Conventional sorting and inspection machines are often designed for specific component types or inspection parameters, making them inflexible to accommodate part variations or design updates. When new models or parts are introduced, existing systems require complete redesign or replacement, leading to increased downtime, high operational costs, and reduced production efficiency.

To address these limitations, this project proposes the design and simulation of a reconfigurable sorting and inspection machine capable of adapting to different automotive components and quality inspection requirements. The reconfigurable design concept aims to enable rapid adjustment of sensors, conveyors, and inspection modules to handle multiple part families without extensive mechanical or software reconfiguration. This flexibility enhances the system's adaptability to dynamic production environments and supports the automotive industry's demand for modular, scalable, and cost-efficient solutions.

The Computer-Aided Design (CAD) of the proposed machine will be developed using advanced CAD tools to model various reconfiguration scenarios for part sorting and defect detection. Simulation studies will be performed to evaluate the system's sorting accuracy, inspection speed, and reconfiguration time under different operational conditions.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of a Reconfigurable Moulding Tool for Complex Part Geometries

Reference	: BR6
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Conventional moulding tools are typically designed for a single product geometry, making them inflexible and costly to modify when part designs change. In industries where product variety and customization are increasingly demanded, dedicated moulds lead to high retooling costs, increased downtime, and wasted material resources. Each new part geometry often requires an entirely new mould, which limits production agility and delays product development cycles.

To address these challenges, this project proposes the design and simulation of a Reconfigurable Moulding Tool capable of adapting to complex and varying part geometries with minimal retooling effort. The system will incorporate modular and adjustable components that can be reconfigured to suit different shapes, sizes, and contours of moulded parts. This flexibility enhances the responsiveness of manufacturing systems, allowing for efficient production of multiple product variants using a single adaptable tool.

The Computer-Aided Design (CAD) of the reconfigurable moulding tool will be developed using advanced CAD software to simulate its functionality and evaluate its ability to form various part families. The tool's modular design will integrate reconfigurable inserts, adjustable cavity sections, and quick-change mechanisms to accommodate complex geometries without compromising precision or surface quality.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Optimisation and Performance Evaluation of a Semi-Automated Reconfigurable Welding Jig for Automotive Component Manufacturing

Reference	: BR7
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Welding is a critical process in automotive component manufacturing, where high dimensional accuracy, repeatability, and production flexibility are essential. Traditional welding jigs, however, are typically dedicated fixtures designed for a single component geometry. This lack of flexibility becomes a major limitation in modern automotive environments where frequent design updates, model variations, and low-volume/high-mix production demand adaptable fixturing solutions. To address these challenges, this project aims to optimise and evaluate a semi-automated reconfigurable welding jig tailored for diverse automotive components. The jig integrates adjustable locating pins, modular clamping units, and sliding fixture plates to allow operators to reconfigure the setup with minimal tooling and reduced downtime. Furthermore, the jig has semi-automation features, including sensors, pneumatic or electric clamping mechanisms as well as positioning and safety systems. A control subsystem is integrated to manage key functions such as clamp actuation, component positioning feedback, and sequence control. MATLAB/Simulink will be used to optimise and validate the control logic, particularly for clamping timing, position sensing, and safety interlocks. The project will involve the fabrication of a fully functional prototype of the reconfigurable welding jig. Performance evaluation will then be conducted by testing the jig with different automotive component geometries, assessing parameters such as setup time, repositioning accuracy, clamping repeatability, weld quality consistency, and operator usability. A simulation model will also be developed to demonstrate the jig's capability to handle various component designs and welding configurations.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Machine Design A (314) | Mechatronics (424)

Design and Integration of a Load-Cell-Based Force Monitoring System for a Modular Railway Wheel-Set Test Rig

Reference	: BR8
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project focuses on enhancing a modular, small-scale railway wheel-set test rig through the design and integration of a load-cell-based force monitoring system. The original rig, developed to simulate wheel–rail interaction under controlled vibration, incorporates mechanical, electrical, and data acquisition subsystems that enable realistic dynamic testing of a railway wheel-set. While the prototype roller rig demonstrated stable actuator motion, smooth rotational behaviour, and vibration responses that aligned with theoretical expectations, the next advancement required accurate measurement of contact forces between the wheel and rollers. The objective of this project is to design, implement, and calibrate a load-cell-based sensing system capable of capturing vertical and tangential forces generated during wheel–roller interaction. The system is integrated into the existing modular structure without altering its core mechanical layout, ensuring compatibility with future upgrades such as metal roller assemblies and closed-loop vibration control. The force-monitoring module includes load-cell selection and placement, signal conditioning circuitry, data acquisition integration, and real-time force visualisation. Testing and validation activities involve applying controlled actuator inputs and comparing measured forces with expected values derived from vibration and contact-mechanics models.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Machine Design A (314) | Mechatronics (424)

Thermal comfort in houses

Reference	: WJS5
Supervisor	: Dr Willie Smit
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The student will design and build sensors to be installed in several houses within the same suburb. The objective is to measure the thermal comfort of these houses and investigate how various factors, such as building materials, window sizes, and floor plans, impact this comfort.

The student will conduct thermal calculations or simulations for each house and then compare the theoretical results with actual measurements. The project aims to highlight two key points: 1) the accuracy of the simulations or calculations used by architects, and 2) the variations in thermal comfort resulting from different building materials and house designs.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Mechatronics (424)

Hospital based robotic assistant (competition)

Reference	: WSWART1
Supervisor	: Mr Wayne Swart
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university. You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive. Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Mechatronics (424) | Modelling (334)

Sumo robot (Competition)

Reference	: WSWART5
Supervisor	: Mr Wayne Swart
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames (<https://robogames.net/rules/all-sumo.php>) international sumo robot competition rules for the Mini Sumo robot class. Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.

Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electronics (245) | Machine Design A (314)

Digital Twin Launchpad: Develop Next-Generation Engineering Tools

Reference	: NCT1
Supervisor	: Dr Nicole Taylor
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Are you excited by cutting-edge engineering technologies, such as digital twins, data pipelines, and human-system interaction? Do you want to graduate with skills that are increasingly in demand in modern engineering and research environments? If so, this project may be an ideal fit for you.

This final year project forms part of a larger research initiative to develop the Digital Twin Launchpad - a human-centred, modular platform designed to accelerate the creation of digital twin solutions. The Launchpad will serve as a toolbox of software components that make digital twin development faster, more intuitive, and more accessible for engineers and researchers, particularly those who may not be specialists in data engineering.

The overarching research goals align with the principles of Industry 5.0, emphasising human-centred design and the integration of human factors into digital environments. The platform aims to:

- Support the creation of digital twins that not only mirror the physical world but also act as research instruments to automate and advance scientific enquiry.
- Provide researchers with ready-to-use tools that minimise the steep learning curve associated with complex data engineering practices.
- Enable richer analysis of human factors by embedding human-centred thinking into digital twin development workflows.

Your contribution to the project will require you to select a component of the Digital Twin Launchpad to design and develop. Available project focus areas include:

- Data pipeline constructor tools
- Data ingestion frameworks
- Data model design and hosting tools
- Data visualisation modules
- User interface for the Digital Twin Launchpad
- Research data management and maintenance tools

Your developed component will be showcased through a case study implementation involving human digital twins aboard the SA Agulhas II, South Africa's polar supply and research vessel. This case study should support the development of tools to analyse human factors at sea - including motion sickness, discomfort, and sleep disturbance - offering a rare opportunity to contribute to real-world engineering research with meaningful human impact.

By completing this project, you will develop:

- Hands-on experience with state-of-the-art digital twin technologies
- Skills in data engineering, modelling, visualisation, and developing software architectures
- Experience designing human-centred engineering systems
- Exposure to real research datasets
- A portfolio-ready project demonstrating innovation and technical depth

This project is ideal for students eager to work at the frontier of modern engineering and make a tangible contribution to an emerging research platform used in multidisciplinary contexts.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Develop a Modular Mine-Like Testbed for Next-Generation Mining Technology

Reference	: NCT2
Supervisor	: Dr Nicole Taylor
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Are you interested in cutting-edge engineering technologies, such as smart mining systems, human-centred sensing, and Industry 4.0/5.0 automation? Do you want to graduate with hands-on experience in designing reconfigurable, real-world engineering systems? If so, this project may be ideal for you. This final-year project forms part of a broader research initiative to develop a mobile mine testbed - a human-centred, Industry 5.0-aligned environment for testing next-generation mining technologies. The testbed will serve as a mobile simulation of mine-like settings, such as underground mine tunnels, enabling safe, realistic evaluation and showcasing of sensors, wearables, communication networks, and distributed technology systems. It will also complement the existing SIMLAB mining simulation facility (<https://sites.google.com/view/mad-research-group/facilities/simlab?authuser=0>), which is temporarily unavailable due to ongoing renovations.

The overarching research goals aim to emphasise human, environment and system integration in mine-like settings. The testbed aims to:

- Provide a realistic, modular environment for validating smart-mining technologies.
- Support the integration of human-centred devices, such as a previously developed wearable human-condition monitoring device for mineworkers.
- Enable plug-and-play testing of sensors, communication systems, and distributed computing infrastructures.

Your contribution to the project will require you to design and develop components of the testbed, which include focus areas:

- Modular structural design that mimics mine-like settings, such as underground mine tunnels
- Integration of environmental and human-centred sensors
- Implementation of adaptable communication and networking systems
- Development of data acquisition and analysis through computing technologies
- Validation and showcasing of plug-and-play operation for a case study

By completing this project, you will develop:

- Hands-on experience in modular systems design
- Skills in sensor integration, communication networks, and embedded technologies
- Experience with human-centred engineering concepts in Industry 4.0/5.0 contexts
- Exposure to real mining-research environments and technologies
- A portfolio-ready project that showcases innovation and practical engineering ability

This project is ideal for students who want to work at the frontier of smart-mining technologies and contribute to a long-term research platform used across multiple engineering disciplines.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Mechatronics (424) | Experimental Techniques (414)

Rough Seas, Real Humans: Develop a Tool to Analyse Motion Sickness, Discomfort and Sleep Disturbance

Reference	: NCT3
Supervisor	: Dr Nicole Taylor
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Are you curious about how real people experience the harsh conditions of life at sea - particularly the vibration, slamming, and rigid-body motion that can cause discomfort, sleep disturbance, and motion sickness? Do you want to apply your engineering skills to a project with direct human impact and real-world data from a polar research vessel? If so, this project could be the perfect fit. This final-year project focusses on developing a human-centred analysis tool for studying how ship motion affects people aboard the SA Agulhas II, South Africa's polar supply and research vessel. The goal is to transform paper-based and digital datasets into an automated system for analysing human factors related to:

- Ship slamming events and comfort effects
- Rigid-body motion linked to motion sickness
- Sleep disturbance

Your project will support ongoing research into improving human well-being and operational decision-making during scientific voyages in rough sea conditions (<https://svrg.sun.ac.za/>). Your contribution to the research will include:

- Digitise and structure historical paper-bound questionnaires from previous voyages.
- Develop algorithms to analyse slamming, motion sickness, and/or sleep disturbance data.
- Build an integrated analysis tool that makes these datasets easy to explore and interpret.
- Prototype visualisations or dashboards that reveal insights about human experiences at sea.

By completing this project, you will develop:

- Experience with data cleaning, digitisation, and structuring
- Skills in algorithm development and time-series analysis/signal processing of real-world data collected on the SA Agulhas II during previous voyages
- Understanding of human factors engineering in extreme environments
- A polished, portfolio-ready tool used in ongoing multidisciplinary research

This project is an ideal choice for students who enjoy working with data, solving meaningful engineering problems, and building tools that directly serve people at sea in demanding real-world environments.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Automatic Wheel Choker Design for Industrial Vehicles

Reference	: LZ1
Supervisor	: Prof Lijun Zhang
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

• Structural design and simulation of a wheel choker. • Adaptable to different terrain conditions. • Adaptable to different vehicle dimensions. • Structural decoupling from the rigid body of the vehicle. • Mechatronics students only.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electronics (245) | Electronics (315)

1.2 Mechanics Division

Improved measurement of powder spreadability for additive manufacturing

Reference	: DB1
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Laser powder bed fusion (LPBF) is an additive manufacturing technique that uses a high power laser to fuse thinly spread layers of metal powder together, to progressively build the required part shape, layer by layer. Metal powders consist of a range of powder particle shapes and sizes which causes them to pack, flow and spread in different ways. Powder characterisation techniques attempt to provide a set of parameters that will help engineers in designing powder manufacturing processes that produce high quality, defect-free parts. For LPBF, powder spreadability is an important parameter in ensuring consistent powder layers and final material quality. SU's Materials laboratory has a custom-built powder spreadability test rig that was designed to help characterise powders in terms of their suitability for LPBF processing. A previous project designed a wedge-system to allow for fine adjustment (10 μm) of the layer height for spreadability testing; however, the accuracy of this adjustment is negatively affected by the stacking errors of tolerance in the system and requires refinement. The goal of this project is to implement design adjustments to allow for more accurate and repeatable spreadability testing using the wedge height adjustment mechanism. As such, it will entail a design review, new design implementation, testing of the design changes through evaluation of the spreadability of commonly used AM powders in the laboratory.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Experimental Techniques (414)

Evaluation of a flash apparatus for measuring thermal conductivity of metal powder structures

Reference	: DB2
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

The laser flash method is an advanced application of the original flash method, developed in 1961 by Parker et al, to measure the thermal conductivity of materials. Commercially available laser flash equipment is expensive and has to be imported into South Africa. A low-cost version of this apparatus, using an LED source for heating, was designed and built in 2025, and initial experiments showed that it yields promising results. The aim of this project is to review the design, implement design changes to improve the measurements, and use the apparatus to measure the thermal conductivity of loose metal powder and sintered structures.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Heat Transfer A (414) | Machine Design A (314)

Metal powder screen printing apparatus: design improvement and evaluation

Reference	: DB3
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

A laboratory scale 3D screen printing apparatus has been designed and built for manufacturing thin metal structures from metal powder slurries. While the screen printing mechanism works well, the equipment has yet to produce a cohesive thin metal film as required. The goal of this project is to investigate different slurry formulations and screen printing parameters with the purpose of printing a cohesive thin metal film. Additionally, the process requires a heating stage for drying the slurry after deposition. The heating mechanism for the build plate must be designed and integrated into the existing system. The design must be evaluated for defect-free drying of the thin films after slurry deposition.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Heat Transfer A (414) | Experimental Techniques (414)

Sacrificial anode depletion detection for water heater

Reference	: DB7
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

US Patent US20140218005A1 was abandoned in 2016 which means it is now freely available for general use. The patent describes a hardware circuit for measuring the current between a sacrificial anode and a water heater shell for monitoring anode depletion (<https://patents.google.com/patent/US20140218005A1/en>). Sacrificial anodes are instrumental in ensuring extended life of water heaters (geysers) but visual checking of them is very poorly monitored in most households as it requires removing and visually inspecting the anode on a regular basis. With this design, monitoring the anode rod current allows for remote monitoring of the system and can provide a means of indicating when the anode needs to be replaced. The goal of this project is to design a process for retrofitting locally available sacrificial anodes with the anode current detection device, evaluating the setup, and identifying a solution that would be viable for commercialisation in South Africa.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Design (E) (314) | Experimental Techniques (414)

Corrosion solutions for 18650 lithium-ion rechargeable batteries

Reference	: DB8
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

18650 Lithium-ion batteries are a standardised and widely-used type of cylindrical rechargeable battery, utilised over a wide range of consumer products such as torches, laptops, electric vehicles and power tools. A common issue that arises in apparatus that use these batteries is corrosion of the positive terminal. The goal of this project is to investigate potential causes for this corrosion, and to evaluate different approaches to counteracting this corrosion. Currently, aluminium foil tape is used as a sacrificial anode, or contact grease as a protective coating. A full investigation into the mechanisms that govern these interventions, and a description of how each solution acts to prevent corrosion is required, along with recommendations for best use case scenarios.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Mechatronics (424) | Experimental Techniques (414)

Vibration-Assisted Powder Flow Optimization For Multi-Material recoater System

Reference	: RB1
Supervisor	: Mr Raymond Botete
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Laser Powder Bed Fusion (LPBF) is a leading additive manufacturing process for producing high-precision metal components. Despite its advantages, incorporating multiple materials within a single build remains a significant challenge due to difficulties in accurate and selective powder deposition. A previous project developed an adaptive powder recoater capable of spreading multiple powders on the same build platform, but testing revealed critical limitations in powder flow, layer accuracy, and material integrity. To address these issues, this project focuses on enhancing the nozzle design by integrating a piezoelectric vibration system to improve powder flow and reduce clumping. The student will experimentally determine optimal vibration frequencies and amplitudes for different metal powders and validate their effect on achieving uniform deposition across varying layer heights and spreading patterns. Deliverables include a functional vibration-enabled nozzle assembly, a dataset correlating vibration parameters with deposition quality, and recommendations for operating conditions to ensure consistent multi-material spreading.

Key modules (Scientific and Engineering Knowledge):

Vibration and Noise (354) | Experimental Techniques (414)

Automated Powder Deposition Control For Multi-Material Recoater System

Reference	: RB2
Supervisor	: Mr Raymond Botete
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Accurate powder deposition in multi-material LPBF requires precise coordination of nozzle actuation, valve timing, and powder flow. The previous adaptive powder recoater prototype experienced start/stop deposition artifacts and unreliable valve performance due to limited electronics and firmware capabilities. This project aims to upgrade the control system, implementing advanced timing logic to synchronize solenoid valve actuation with vibration and nozzle motion, reducing deposition errors and improving repeatability.

The student will develop, integrate, and test the upgraded control system on the existing prototype, evaluating improvements in deposition consistency and reliability. Deliverables include a fully functional automated control system, test results demonstrating reduced start/stop artifacts, and operational guidelines for reliable multi-material deposition.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Design (E) (314) | Mechatronics (424)

Contamination Control & Purge Strategies For Multi-Material Recoater System

Reference	: RB3
Supervisor	: Mr Raymond Botete
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Laser Powder Bed Fusion (LPBF) enables high-precision metal fabrication, but using multiple powders in a single build introduces the risk of cross-contamination, compromising layer quality and material integrity. The existing adaptive powder recoater prototype demonstrated the ability to deposit multiple powders, yet residual material on the nozzle and build surface led to cross-contamination between successive powders. This project focuses on designing, developing, and integrating an effective cleanup mechanism into the prototype to remove residual powders between material changes.

The student will be responsible for conceptualising a cleanup solution, implementing it on the prototype, and experimentally validating its effectiveness. Deliverables include a fully integrated, functional cleanup system, experimental data demonstrating reduced cross-contamination, and operational guidelines for reliable multi-material deposition in LPBF processes.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Experimental Techniques (414)

Tabletop dynamic linear fatigue tester

Reference	: NMD1
Supervisor	: Dr Nur Mohamed Dhansay
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

This project requires the design, build and testing of a small scale (tabletop) fatigue tester. Ideally, it needs to house a compact tension specimen and conform to the ASTM E647 standard. The fatigue tester will have to control and monitor the load and frequency of the dynamic cycles in various ways. The interest lies in that the student will have to incorporate a (or multiple) suitable crack detection method(s) which can automatically adjust the load level accordingly (variable amplitude loading as opposed to constant amplitude loading). This means that the fatigue tester should also be capable of being controlled via the stress intensity factor. Given that the system will have linear motion, suitable alignment technique needs to be in place to assure that specimen alignment is accurate and does not cause any significant/unwanted bending. This tester should also incorporate a camera attachment for both remote viewing and control.

It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.

For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Small scale rolling contact fatigue tester for railway track

Reference	: NMD2
Supervisor	: Dr Nur Mohamed Dhansay
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Rolling contact fatigue (RCF), as the name suggests, is a near surface fatigue process which occurs between surfaces which are in contact with each other. This phenomenon is typically found in bearings, gears, railway lines etc. The issue lies in that testing RCF for railways, typically, occur at full scale. This of course requires a large space and resources to move the test specimens (railway line and wheelset) to the relevant location.

The student is required to design, build and test a small scale RCF tester specific to railway tests. This means that the equipment needs to consider straight and curved rolling contact regimes, inner and outer styled wheel flanges, mapping of the lateral force and angle of attack, amongst others, to analyse the wear effects on the wheelset and railway track.

It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.

For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Rotating bending fatigue machine

Reference	: NMD3
Supervisor	: Dr Nur Mohamed Dhansay
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

This project requires the design, build and testing of a rotating bending fatigue tester. Ideally, it needs to conform to the ASTM E2948 and/or ISO1143. The fatigue tester will have to control and monitor the load(s) and frequency of the dynamic cycles in various ways. The system should also be capable of being controlled and monitored remotely. This type of testing allows for fully reversed, uniaxial type, fatigue loading using rotation rather than linear actuation. It is highly desired for the student to add torsional fatigue to the system, resulting in a multiaxial type fatigue testing machine. It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.

For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Improvement on the design of a rotating drum mechanism for measuring dynamic angle of repose

Reference	: GOV1
Supervisor	: Mr Preyin Govender
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The dynamic angle of repose refers to the maximum angle at which a material can be piled or stacked without the particles sliding or cascading off. Unlike the static angle of repose, which is determined when the material is at rest, the dynamic angle takes into account the effects of motion or flow. It is commonly observed in powders and is influenced by factors such as particle shape, size distribution, and moisture content. The dynamic angle of repose is important in industries like material handling, where the behavior of bulk materials during transportation or storage must be understood and managed. The student will be tasked with improving the design and manufacturing of a cost friendly rotating drum that was used to determine the dynamic angle of repose for various metal powders.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Mechatronics (424)

The Effect of Powder Size Distribution (PSD) of WC-Co powders on Build Quality through laser powder bed fusion

Reference	: GOV2
Supervisor	: Mr Preyin Govender
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

The purpose of this investigation analyses how the powder size distribution (PSD) of WC-Co hardmetal powders influences powder spreading, porosity formation, and microcracking during Laser Powder Bed Fusion (LPBF). WC-Co is a challenging material to process through additive manufacturing due to its high melting point, low ductility, and susceptibility to thermal cracking. The PSD strongly affects powder flowability, packing density, and energy absorption, making it a key parameter in achieving high-quality LPBF builds.

In this study, the student will be tasked with preparing various powder sizes by sieving existing WC-Co powder. Each batch will be characterised for particle morphology and flowability. Small LPBF test cubes will be printed using identical process parameters to isolate the effect of PSD. Microstructural analysis will quantify porosity, crack density, and final hardness. The results will help determine which PSD provides the best balance of spreadability and thermal behaviour for printing dense, crack-minimised WC-Co components. This work supports the optimisation of hardmetals processing for wear applications in mining and tooling.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Strength of Materials W (334)

Laser Re-melting Effect on WC-Co Manufactured via Laser Powder Bed Fusion

Reference	: GOV3
Supervisor	: Mr Preyin Govender
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

The aim of this investigation is to explore how in-situ laser re-melting influences the surface quality, microstructure, and crack formation of WC-Co hardmetals produced by Laser Powder Bed Fusion (LPBF). WC-Co commonly suffers from high thermal gradients during printing, leading to microcracking, binder (Co) depletion, and poor surface finishes. Laser re-melting, where the laser scans the top layer again after deposition, may help reduce surface roughness, heal shallow defects, and improve microstructural homogeneity.

In this study, small WC-Co samples will be manufactured using identical base parameters, but with varying degrees of re-melting. Each condition will be evaluated for changes in surface roughness, porosity, and crack density. Microhardness testing will assess whether re-melting alters carbide distribution or binder pooling. The outcomes will help determine whether re-melting is an effective, low-cost strategy to enhance build quality and reduce defect formation in LPBF-produced WC-Co components, supporting improved performance for wear-resistant applications.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Strength of Materials W (334)

Microcracking Analysis of WC-Co Using Various Printing Parameters in Laser Powder Bed Fusion

Reference	: GOV4
Supervisor	: Mr Preyin Govender
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

WC-Co is a hard-metal composite where tungsten carbide particles provide extreme hardness and wear resistance, while the cobalt binder gives toughness. It is widely used in cutting tools, drilling bits, wear-resistant components, mining equipment, and additive-manufactured hard-metal parts. This project examines how various printing parameters influence microcrack formation in WC-Co parts manufactured using Laser Powder Bed Fusion (LPBF). WC-Co hardmetals are prone to severe thermal stresses during printing, often resulting in microcracks that reduce strength and reliability. Understanding how process parameters control these defects is essential for improving printability and material performance.

The study will print a series of small LPBF samples using combinations of different printing parameter sets with varying energy density. Each sample will be analysed for crack density, porosity, and microhardness, providing a comprehensive view of how thermal gradients and melt pool behaviour contribute to cracking. The results will identify parameter combinations that minimize microcracking and optimise density, offering valuable insight for producing high-quality WC-Co components through LPBF.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Strength of Materials W (334)

Design, build, and test a classroom demonstration tool for the planar mechanisms presented in Introductory Machine Design 254

Reference	: VS2
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

A key concept in the second-year module, Introductory Machine Design 254, is the understanding of the kinematics of planar mechanisms. The aim of this project is to develop a series of class demonstration tools to help students with the visualisation of the different planar mechanisms studied in the course. These tools should be adjustable to allow for different motions to be obtained. Real-time imaging analysis should be used to determine the position, velocity, and acceleration of the mechanisms and code should be developed to compare the analytical solutions to the demonstration solutions.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Modelling (334) | Vibration and Noise (354)

Design, build, and test a balancing rig for classroom practical demonstration in Introductory Machine Design 254

Reference	: VS3
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Rotating unbalance and shaft balancing is a topic presented in the second-year module, Introductory Machine Design 254. Stellenbosch University does not currently have a set up to demonstrate the effects of shaft unbalance in a classroom/practical environment. While the effect of an unbalanced rotating shaft on the bearing loads can be shown via video examples, the built device will allow students to develop an analytical solution for balancing the rotating shaft and then implement their solution and observe the results on the loads transferred to the bearings and the structure attached to the rotating shaft.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Modelling (334) | Vibration and Noise (354)

Numerically determine the rupture threshold for an armor steel to projectile impact

Reference	: VS4
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: Finite Element Methods (FEM)

Description:

The goal of this project is to use Finite Element Analysis (FEA) techniques to simulate the dynamic interaction between a projectile and an armor steel target plate to determine the rupture threshold of plate. The study aims to provide insights into the material's dynamic structural response and determine the key failure mechanisms. The project will involve the selection of a suitable armor steel and the characterization of the material selected. A 3D numerical model will need to be developed including the selection of appropriate element formulations and material models. The numerical model results must be validated against appropriate published data.

In this project, you will be required to learn a new FEA tool, namely, LS-DYNA. This software will be used for the dynamic simulations on the steel specimens. You will also need to choose your material and design your material tests during the first semester as this data is essential for characterising the material models in the FEA software. Thus, you will need to learn about the existing material models and the material tests required to characterise these models.

Key modules (Scientific and Engineering Knowledge):

Finite Element Methods (414) | Strength of Materials W (334) | Experimental Techniques (414)

Redesign of an extendable threshold to close the gap on platforms

Reference	: VS5
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

To improve the accessibility of train carriages, there is a proposal to redesign the thresholds of the trains to close the gap between the train and the platform edge. This will allow people with mobility issues to more easily board and exit the train carriages, and also allow items such as heavy suitcases to be loaded. The design must be capable of withstanding repeated impact loads from passengers and luggage, while also being able to operate in a range of environmental conditions such as rain and high temperatures. It needs to incorporate sensors to locate the edge of the platform and motors to extend the platform until the gap is closed. This project is being run by the Gibela research group.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Investigation of the effects of out-of-autoclave curing on the performance of composite specimens.

Reference	: VS7
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Composite materials are extensively used in aerospace, automotive, and defence applications due to their excellent properties, such as corrosion resistance and high strength-to-weight ratio and their ability to be used to manufacture components with complex shapes and optimised stiffness and deformation characteristics. Out-of-autoclave curing serves as an alternative and cheaper option for the manufacture of composite specimens due to the lower comparative costs of large volume temperature-controlled ovens.

In this project, at least two fibre-reinforced matrix materials should be considered in combination with Nomex honeycomb core moulded in a sandwich format. The primary aim is to investigate the strength properties of the cured composite specimens manufactured under different curing conditions (autoclave vs oven). The specimens would need to be manufactured at AAT Composites in 2026 due to the unavailability of the oven. This project requires direct participation with an industrial partner at their facilities and will require careful time management and early engagement with AAT Composites. This project has been proposed by AAT Composites, who will be able to assist with raw materials and provide technical assistance on layup configurations to investigate.

Key modules (Scientific and Engineering Knowledge):

Strength of Materials W (334) | Experimental Techniques (414)

Composite specimen material testing and material model characterisation

Reference	: VS8
Supervisor	: Dr Vinay Shekhar
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Finite element (FE) models are useful for the rapid design iteration and optimisation of composite parts for under different loading conditions that can be difficult to replicate easily and quickly in a laboratory testing environment. The accuracy of the FE simulations is highly dependent on the accuracy of the material data for the composite materials used. Thus, there is a need to characterise these materials in accordance with testing standards. This project requires direct participation with an industrial partner at their facilities and will require careful time management and early engagement with AAT Composites.

This project aims to characterise the mechanical behaviour of woven fibre-reinforced polymer (FRP) composite specimens manufactured and supplied by AAT Composites. Material testing will be performed based on appropriate testing standards and utilising non-contact displacement measurement techniques such as Digital Image Correlation (DIC). The data collected will need to be fitted to suitable material models and validated against simple FE models. The data generated will aid the numerical analysis team at AAT Composites in the development of FE models for the components being developed at their facilities.

Key modules (Scientific and Engineering Knowledge):

Finite Element Methods (414) | Strength of Materials W (334) | Experimental Techniques (414)

1.3 ThermoFluids Division

Development of a 1D model of a residential-scale air-to-water transcritical CO₂ heat pump with an internal heat exchanger

Reference	: RAH1
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO₂ (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO₂ (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO₂ temperatures in the supercritical region. The integration of an internal heat exchanger may improve system efficiency due to the higher gas cooler outlet temperatures.

The aim of this project is to develop two simplified component-level (1D) models of a residential-scale transcritical CO₂ heat pump, (i) with an internal heat exchanger, and (ii) without an internal heat exchanger. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, (dry) air-source evaporator, and internal heat exchanger. The internal heat exchanger will be sized and modelled in detail. The two developed models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to evaluate the effects of an internal heat exchanger, and to quantify the performance of the heat pump system at typical ambient conditions observed in Cape Town.

Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of an industrial-scale air-to-water transcritical CO₂ heat pump with an internal heat exchanger

Reference	: RAH2
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO₂ (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO₂ (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO₂ temperatures in the supercritical region. The integration of an internal heat exchanger may improve system efficiency due to the higher gas cooler outlet temperatures.

The aim of this project is to develop two simplified component-level (1D) models of an industrial-scale transcritical CO₂ heat pump, one with and another without an internal heat exchanger. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, (dry) air-source evaporator, and internal heat exchanger. The internal heat exchanger will be sized and modelled in detail. The two developed models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to evaluate the effects of an internal heat exchanger, and to quantify the performance of the heat pump system at typical ambient conditions observed in Joburg.

Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Performance comparison of a residential-scale air-to-water conventional heat pump and transcritical CO₂ heat pump through 1D simulation

Reference	: RAH3
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO₂ (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO₂ (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO₂ temperatures in the supercritical region. However, this requires high operating pressures which can result in large losses across the expansion valve compared to conventional heat pump systems.

The aim of the project is to develop two simplified component-level (1D) heat pump models: (i) of a conventional heat pump system, and (ii) of a transcritical CO₂ heat pump system, for residential-scale applications. The models must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to quantify performance differences and feasibility of the two systems with respect to ambient conditions in Joburg.

Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Performance comparison of an industrial-scale air-to-water conventional heat pump and transcritical CO₂ heat pump through 1D simulation

Reference	: RAH4
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO₂ (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO₂ (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO₂ temperatures in the supercritical region. However, this requires high operating pressures which can result in large losses across the expansion valve compared to conventional heat pump systems.

The aim of the project is to develop two simplified component-level (1D) heat pump models: (i) of a conventional heat pump system, and (ii) of a transcritical CO₂ heat pump system, for industrial-scale applications. The models must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to quantify performance differences and feasibility of the two systems with respect to ambient conditions in Cape Town.

Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of a transcritical CO₂ heat pump in Flownex SE for performance characterization in typical South African ambient conditions

Reference	: RAH5
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO₂ (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO₂ (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO₂ temperatures in the supercritical region, but requiring high operating pressures.

The aim of this project is to develop a 1-D model of a transcritical CO₂ heat pump in Flownex SE. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The developed model will be verified using existing results from either literature or from an existing model. Once developed and verified, the model will be used to characterize performance of the heat pump system for average South African ambient conditions observed on a yearly basis. Transient modelling on a daily-basis may also be explored.

The candidate should have a strong interest in thermofluids and simulation. Knowledge of C# scripting may be beneficial to develop custom scripts in Flownex SE, but is not required.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of a simple recuperated supercritical CO₂ Brayton cycle in Flownex SE to investigate nominal and part-load operation

Reference	: RAH6
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Supercritical CO₂ (sCO₂) Brayton power cycles have garnered significant research attention for its high thermal efficiency and compact physical footprint compared to traditional steam Rankine cycles. This makes sCO₂ Brayton cycles highly suitable to use as a topping/bottoming cycle to retrofit existing power cycles, thereby increasing power generation and energy efficiency.

While there are several sCO₂ Brayton cycle layouts, this project focuses on a simple recuperated Brayton cycle layout, applied to retrofit an existing Rankine cycle. The candidate will develop a 1D model of the cycle in Flownex SE, capturing cycle component characteristics at a high level. The recuperator can have a significant impact on cycle performance. Thus, detailed recuperator modelling will be integrated into the model. The developed model will be verified at nominal load. Thereafter, the model will be transiently simulated to investigate system performance and response across load changes.

The candidate should have a strong interest in thermofluids and simulation. Knowledge of C# scripting may be beneficial to develop custom scripts in Flownex SE, but is not required.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Investigation on the operation of a simple recuperated sCO₂ Brayton power cycle near the critical point using 1D simulation

Reference	: RAH7
Supervisor	: Dr Rashid Haffeejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Supercritical CO₂ (sCO₂) Brayton power cycles have garnered significant research attention for high thermal efficiencies and compact physical footprints compared to traditional steam Rankine cycles. This makes sCO₂ Brayton cycles highly suitable to use as a topping/bottoming cycle to retrofit existing power cycles, thereby increasing power generation and energy efficiency. The critical point poses several challenges with cycle operation. Rapid fluctuations in thermo-physical properties of sCO₂ near the critical point can significantly affect both compressor behaviour (due to density changes) as well as pre-cooler behaviour (due to specific heat changes). These variations are further exacerbated with load fluctuations influencing heat loads and cycle operation.

This project focuses on a simple recuperated Brayton cycle layout, applied to retrofit an existing Rankine cycle. The candidate will develop a 1D model of the cycle in Python, capturing cycle component characteristics at a high level. The developed model will be verified using a previously designed set of nominal cycle parameters. The model will then be used to explore two sets of cycle operating parameters: (i) low-side pressure and temperature away from the pseudo-critical point (as previously designed), and (ii) low-side pressure and temperature closer to the critical point. This may require re-design of each cycle component to suit the developed operating parameter specifications. The consequent designed cycles will be investigated for various load conditions, which will characterise cycle performance across the operating range.

Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Hospital based robotic assistant (competition)

Reference	: RAH8
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university.

You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.

Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electronics (315) | Mechatronics (424)

Sumo robot (Competition)

Reference	: RAH9
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames international sumo robot competition rules for the Mini Sumo robot class, see: <https://robogames.net/rules/all-sumo.php>. Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.

Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Design (E) (314) | Modelling (334)

Thermal radiation extinction coefficient in a packed bed

Reference	: JEH2
Supervisor	: Prof Jaap Hoffmann
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

A test facility to determine the extinction coefficient in a packed bed was developed in 2025. The power supply was insufficient to heat the bed to temperatures where thermal radiation dominates. This project is aimed at revising the heating and measurement system. A packed bed can be approximated as a participating medium in radiation calculations. This allows easy implementation of radiation heat transfer in a computational fluid dynamics model. Such a model requires representative values of the absorption and scattering coefficients in the bed. The candidate must develop a mathematical model to capture the radiation phenomena, and then extract the relevant radiation properties of the bed experimentally. It is conceivable that the radiation effect can be measured in a bed with a heated top, cooled bottom and insulated sides.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Experimental Techniques (414)

Particle image velocimetry in a randomly packed bed of ellipsoidal particles

Reference	: JEH3
Supervisor	: Prof Jaap Hoffmann
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.

Transition to turbulence in packed beds happen at Reynolds numbers of about 350 (depending of the definition), and most of the available work has been done with beds of regular spaced glass or Perspex rods or direct numerical (CFD) simulation. The former is not representative of a randomly packed bed, whilst the latter requires huge computational resources.

An intern from Eindhoven University of Technology developed a PIV system for a structured bed of ellipsoidal particles. Techniques and materials in manufacturing transparent particles have been developed during the internship. This study aims to proceed with the PIV in a moving fluid.

A small wind tunnel with transparent walls that can be filled with transparent particles immersed in a liquid with a matching refractive index will make the flow in the voids visible. Some discrete element modelling might be useful to determine particle positions and orientations. Other techniques may be used to obtain the position and orientation of the particles.

Specialist skills are required to machine and polish the particles; the technology is available in the Western Cape.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Determine the heat transfer coefficient for an irregular particle

Reference	: JEH4
Supervisor	: Prof Jaap Hoffmann
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.

One way of describing the heat transfer coefficient in a packed bed, is to use data for a single particle and correct it for the presence of neighbouring particles. The objective here is to determine the heat transfer coefficient for a single particle at different Reynolds numbers and different flow angles.

Challenges are heating the particle and measuring surface temperatures with minimum disruption of the flow.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Experimental Techniques (414)

Validate Misdorp's 1D model of heat transfer in a packed bed

Reference	: JEH6
Supervisor	: Prof Jaap Hoffmann
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.

For air/crushed rock thermal energy storage systems, the thermal capacity of the air is so small compared to that of the rocks, that it can safely be ignored. Some researchers suggest that crushed rock is used as a cheap filler material in molten salt systems. Misdorp developed a theoretical model to describe 1D heat transfer in such systems, but his model needs validation.

Develop a test facility, say by using crushed rocks and a liquid, that can provide the necessary data to validate Misdorp's model.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Experimental Techniques (414)

Computational modelling and analysis of the steady-flow haemodynamic environment of bicuspid aortic valves

Reference	: LGK1
Supervisor	: Mrs. Lindi Grobler Kock
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The simplified Bernoulli equation used in clinical practice to estimate the TPG relies on simplifying assumptions which have been shown to both under- and overestimate the TPG depending on the cardiac output (CO) and the type and severity of AS.

In this project, steady-state CFD simulations of blood flowing through bicuspid aortic valve (BAV) pathologies at peak systole will be developed. This project will aim to evaluate the flow environment and pressure gradients across different BAV phenotypes and for a range of COs to determine the performance of the simplified Bernoulli equation in each case.

This project uses StarCCM+ to develop and run the CFD simulations, image processing to calculate the opening area ratio and determine the severity of the stenosis, and fundamental fluid mechanic principles to derive a theoretical secondary loss TPG model.

The student will be expected to largely self-learn the use of StarCCM+ using available documentation, tutorials, and examples, with guidance provided as needed.

Key modules (Scientific and Engineering Knowledge):

Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Experimental design and analysis of the steady-flow pressure gradients across bicuspid aortic valves

Reference	: LGK2
Supervisor	: Mrs. Lindi Grobler Kock
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:

Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The simplified Bernoulli equation used in clinical practice to estimate the TPG relies on simplifying assumptions which have been shown to both under- and overestimate the TPG depending on the cardiac output (CO) and the type and severity of AS.

In this project, the steady-flow pressure gradient across 3D-printed bicuspid aortic valve (BAV) models opened at peak systole will be determined experimentally using a glycerin-based blood analog. This project aims to evaluate the pressure gradients across different BAV phenotypes and for a range of COs, and determine the performance of the simplified Bernoulli equation in each case.

BAV models opened at peak systole will be used. A steady-flow hydraulic test bench will be designed and built to measure the pressure gradient across the BAV models. The experimental results will be compared to an analytical TPG estimation and the standard clinical simplified Bernoulli TPG estimation. This project uses image processing to calculate the opening area ratio and determine the severity of the stenosis, and fundamental fluid mechanic principles to derive a theoretical secondary loss TPG model.

Key modules (Scientific and Engineering Knowledge):

Machine Design A (314) | Experimental Techniques (414)

The effect of aortic root diameter on the pressure recovery in aortic stenosis

Reference	: LGK3
Supervisor	: Mrs. Lindi Grobler Kock
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The effect that the aortic root diameter has on pressure recovery, and therefore the net TPG, is important to consider when evaluating the accuracy of the clinical simplified Bernoulli TPG estimation across different types and severities of AS.

In this project, steady-state CFD simulations will be performed to determine the effect that various aortic root diameters have on the downstream energy dissipation and static pressure recovery for various degrees of calcific and rheumatic AS. Using Star-CCM+, CFD models will be developed for various degrees of calcific and rheumatic AS and various aortic root geometries for a range of physiological flow rates. The TPG will be estimated through post-processing of the simulated results. The simulated results will also be compared to the clinical simplified Bernoulli TPG estimation for each geometric configuration.

The student will be expected to largely self-learn the use of StarCCM+ using available documentation, tutorials, and examples, with guidance provided as needed.

Key modules (Scientific and Engineering Knowledge):

Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Development of a validated tool for determining water flow distribution in a cooling tower spray manifold

Reference	: MO4
Supervisor	: Prof Michael Owen
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

Evaporative cooling towers ranging from large natural draft cooling towers for thermal power plants, to small rooftop mechanical draft cooling towers for HVAC systems, are commonly used in industry. The performance of these cooling towers is highly dependent on a uniform water distribution over the heat exchange media, which is realized by means of set of distribution pipes fitted with relative closely spaced spray nozzles. Due to the proximity of the nozzles, the branch dividing loss coefficient for each nozzle may not be independent. In addition, the volume flow from each branch will be influenced by the volume flow and velocity past each nozzle, reduction in static head due to friction, varying pipe and nozzle diameters along the length of the pipe. The effect of the above variables on the volume flow from each nozzle is not well understood and there is a need for a simplified software tool for analyzing the flow in a distribution pipe for a given set of input parameters.

in this project, the student will use a combination of fundamental principles and numerical simulation (CFD and / or 1D process modelling) to analyze the flow distribution in a cooling tower spray manifold. A validated, user-friendly analytical tool for solving the flow distribution in a distribution pipe under specific conditions must be developed.

Key modules (Scientific and Engineering Knowledge):

Data Analytics (Eng) (344) | Numerical Fluid Dynamics (414)

Investigation on the effect of mixing surfaces in hybrid cooling towers

Reference	: MO5
Supervisor	: Prof Michael Owen
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

The widespread use of artificial intelligence and the ever-increasing demand for cloud data storage has resulted in more and more data centers being constructed worldwide. These data centres are often cooled using an evaporative process that produces a visible plume (cloud) above the cooling towers. Regulations prohibit visible plumes in many locations (notably in Europe) and hybrid cooling towers (a combination of dry and evaporative systems) are used for the purpose of plume abatement.

In a hybrid cooling tower, air is simultaneously drawn through a counterflow wet cooling section and a crossflow air-cooled heat exchanger. These two air streams are then mixed before entering the axial flow fan situated at the outlet of the cooling tower. Mixing surfaces are sometimes used to enhance mixing and plume abatement. The effectiveness and effect of these surfaces on the fan inlet air flow conditions are not well documented. Inlet air flow conditions are a dominant factor affecting fan performance and vibration.

This project involves the sizing of a hybrid cooling tower (using information from literature and thermodynamic calculations) and numerical simulation of a simplified system using computational fluid dynamics. The objective of the work is to quantify the impact of mixing surfaces on plume abatement and on the operating conditions experienced by the axial flow fan.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Simulation of air-cooled condenser performance and associated annual power plant output

Reference	: HP1
Supervisor	: Prof Hannes Pretorius
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Modern thermal power plants in arid and semi-arid locations employ water conserving dry cooling technologies to reject the required heat from the cycle to the environment. Among these technologies are mechanical draft air-cooled condensers (ACCs), which employ a multitude of large diameter axial flow fans to force airflow across heat exchanger bundles.

This topic will involve the thermo-fluid simulation of the performance of an ACC designed for a specific power generation application. By coupling the ACC to a turbine characteristic and using meteorological data for a selected site as input, the energy generation of the power plant will then be determined over a typical year.

The project will include the development of a 0-D ACC simulation model using your choice of non-commercial programming package, and designing an appropriate system size to achieve the required design point performance. Thereafter the model will be used to evaluate the annual performance of the power plant which uses this ACC, for two different heat exchanger finned-tube designs. Site meteorological data will need to be sourced and processed, and the effect of wind will need to be approximated in the model.

As 3 students may select this topic, each will be tasked to size and simulate an ACC for a different power generation application (50 MWe concentrated solar power, 600 MWe (200 MWe steam component) combined cycle gas turbine, or 800 MWe coal-fired application).

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Simulation of natural draft air-cooled condenser performance and associated annual power plant output

Reference	: HP2
Supervisor	: Prof Hannes Pretorius
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Modern thermal power plants in arid and semi-arid locations employ water conserving dry cooling technologies to reject the required heat from the cycle to the environment. Among these technologies are natural draft air-cooled condensers (NDACCs), which employ large dry cooling towers to induce airflow across heat exchanger bundles.

This topic will involve the thermo-fluid simulation of the performance of an NDACC designed for a specific power generation application. By coupling the NDACC to a turbine characteristic and using meteorological data for a selected site as input, the energy generation of the power plant will then be determined over a typical year.

The project will include the development of a 0-D NDACC simulation model using your choice of non-commercial programming package, and designing an appropriate system size to achieve the required design point performance. Thereafter the model will be used to evaluate the annual performance of the power plant which uses this NDACC, for three different heat exchanger finned-tube designs. Site meteorological data will need to be sourced and processed, and the effect of wind will need to be approximated in the model.

As 3 students may select this topic, each will be tasked to size and simulate an NDACC for a different power generation application (50 MWe concentrated solar power, 600 MWe (200 MWe steam component) combined cycle gas turbine, or 800 MWe coal-fired application).

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414)

Hospital based robotic assistant (competition)

Reference	: HP3
Supervisor	: Prof Hannes Pretorius
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office.

Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project.

You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.

Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electronics (315) | Mechatronics (424)

Design and optimisation of a closed loop evaporative cooling tower for liquid cooling

Reference	: HR1
Supervisor	: Prof Hanno Reuter
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

Closed loop evaporative cooling systems are generally used as refrigerant condensers in the HVAC industry. In a specific configuration, these cooling systems can also be used effectively for the cooling of liquids. This project entails the development of a computational and CFD model of such a cooling system and to conduct a parametric study for the optimisation of such cooling systems.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Design and optimisation of a closed loop liquid cooling system with a wet-cooling tower

Reference	: HR2
Supervisor	: Prof Hanno Reuter
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Computational Fluid Dynamics (CFD)

Description:

This project entails the development of a computational model of such a cooling system and to conduct a parametric study for the optimisation of such cooling systems.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Thermofluid design and evaluation of a reverse cycle air conditioning unit for residential space cooling and heating with the aid of simulation

Reference	: PGR1
Supervisor	: Prof Pieter Rousseau
Can be completed in first semester	: No
Students allowed for this topic	: 3
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

The aim of this project is to conduct the thermofluid system design of a reverse cycle wall split air conditioning unit for residential space cooling and heating with a given nominal cooling capacity, and then to demonstrate its year-round performance with the aid of simulation. The design and evaluation will, amongst others, include the following steps: Understanding the layout and operation of existing residential air conditioning systems, analysing the relevant room loads and weather data to determine suitable design requirements, selecting nominal cycle operating parameters, analysing the refrigeration cycle to determine suitable component design specifications, selecting a suitable off-the-shelf compressor, designing suitable condenser and evaporator heat exchanger layouts, sourcing suitable fans, developing process models for each of the components and integrating these into the cycle simulation, applying the integrated simulation to evaluate the year-round performance of the system using hourly weather data. All analyses and models will require programming in Python and there will be no physical experimentation. The project will allow the participation of three students, each independently developing and analysing their system for a different cooling capacity and climatic region, typified by the weather data for Cape Town, Johannesburg, and Durban, respectively. Each student will also evaluate the year-round performance of their system in Uptington to gain insight into the effect of climate variations on their specific design.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Thermofluid Dynamics (344)

Development and simulation of adiabatic cooling system for a sCO₂ Brayton recompression cycle

Reference	: MTC1
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Hybrid dry/wet cooling systems are mainly cooling systems that utilized both dry and wet cooling approaches to address the issue of wet cooling excessive water consumption and dry cooling power consumption in order to enhance the overall power plant performance. This is particularly important in arid regions where water resources are scarce and the dry bulb temperature relatively high making the heat rejection challenging. In this study, an adiabatic dry/wet cooling system is investigated for cooling of a concentrated solar power (CSP) plant using a supercritical carbon dioxide recompression Brayton cycle. The student will make use of a pre-built recompression cycle and, will insert an adiabatic cooling system that can insure a relatively stable compressor inlet temperature. The complete 1-D model will be built in Flownex.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414)

Design and simulation of a deluged bare tubes heat exchanger bundle of a sCO₂ Brayton recompression cycle

Reference	: MTC2
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Hybrid dry/wet cooling systems are mainly cooling systems that utilized both dry and wet cooling approaches to address the issue of wet cooling excessive water consumption and dry cooling power consumption in order to enhance the overall power plant performance. This is particularly important in arid regions where water resources are scarce and the dry bulb temperature relatively high making the heat rejection challenging. In this study, a deluged cooling system is investigated for cooling of a concentrated solar power (CSP) plant using a supercritical carbon dioxide recompression Brayton cycle. The student will make use of a pre-built recompression cycle and, will insert a deluged cooling system that can insure a relatively stable compressor inlet temperature accounting for variable ambient conditions. The complete 1-D model will be built in Flownex.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414)

Design and simulation of shell-and-tubes heat exchanger for sCO₂ Brayton recompression cycle

Reference	: MTC4
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO₂ as working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. Amongst the many cycles sCO₂ configurations known, this study will focus on the Brayton recompression and a full design of a water cooled shell-and-tube heat exchanger is suggested in order to ensure a stable and low compressor inlet temperature (30 and 33°C). The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Modelling (334)

Design and simulation of an air-cooled heat exchanger for sCO₂ Brayton recompression cycle

Reference	: MTC5
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO₂ as a working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. This study will focus on the design and simulation of an air-cooled heat exchanger suggested to operate as sCO₂ pre-cooler and its integration into the Brayton recompression. The sCO₂ outlet temperature is expected to be ranging between 30 - 33°C. The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Modelling (334)

Feasibility study of a sCO₂ recompression cycle cooled by reversed Brayton cycle

Reference	: MTC6
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO₂ as working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. However, challenges of relatively high ambient temperatures recorded in high-solar resources regions (such as Upington) have been identified as major barrier to achieving higher cycles efficiency. In study, the candidate will be required to investigate the possibility of using a refrigeration loop to ensure cooling, and its challenges. The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Design of a natural draft wet cooling tower for concentrated solar powered sCO₂ recompression power block

Reference	: MTC7
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

Traditionally, wet cooling is considered as the primary option for heat rejection in power generation. This is motivated by its ability for effective cooling due to water thermal properties. In this project, effective cooling of the working fluid (sCO₂) is required to ensure overall cycle efficiency. This project targets 30 - 33°C for the carbon dioxide at the outlet of the cooling tower. The candidate is required to build a 1-D simulation model of the cooling tower in Flownex environment and integrate it to an existing sCO₂ Brayton recompression cycle. This project intends to investigate the cycle's overall response variable weather conditions. Water consumption and cycle efficiency are critical for this project, hence the simulation should consider annual profile of the weather conditions in Upington.

Key modules (Scientific and Engineering Knowledge):

Energy Systems M (434) | Heat Transfer A (414)

Design of a Temperature Control Mechanism for an Atlas Copco ZR 630 Compressor to Ensure Reliable Air Supply in Copper Lixiviation Operations

Reference	: MTC8
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Lixiviation, or leaching, is a key hydrometallurgical step in copper processing in which chemical solutions dissolve copper from crushed ore for subsequent recovery. At the Shituru copper plant in the Democratic Republic of Congo, this process relies on a steady supply of compressed air produced by an Atlas Copco ZR 630 oil-free compressor. During summer, the compressor frequently trips due to excessively high discharge air temperatures, exceeding the allowable limit designed to protect downstream rubber piping and connectors from thermal damage. These shutdowns disrupt operations, reduce efficiency, and compromise the stability of the lixiviation circuit. This project requires the student to design an effective control mechanism that maintains the compressor's discharge temperature within a safe operating range under varying ambient conditions. The proposed solution should address thermal loading, improve cooling performance, and ensure continuous, reliable compressor operation to support uninterrupted lixiviation at the Shituru plant.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electrical Drive Systems (324) | Modelling (334)

Design and Simulation of a Redundant Water-Cooled Heat Exchanger System for an Atlas Copco ZR 630 Compressor at the Shituru Copper Plant

Reference	: MTC9
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:

This project focuses on the design and performance simulation (Using Flownex) of a water-cooled heat exchanger system intended to cool the discharge air of an Atlas Copco ZR 630 oil-free compressor operating at the Shituru copper processing plant in the Democratic Republic of Congo. The existing factory-installed coolers regularly experience fouling and reduced heat transfer efficiency, leading to overheating events and operational interruptions during critical plant activities such as copper lixiviation. To improve reliability and ensure uninterrupted air supply during maintenance of the original heat exchangers, the student will design a fully functional redundant water-cooled exchanger sized to handle full compressor load and compatible with existing piping and plant cooling water infrastructure. The project includes thermal design calculations, selection of appropriate heat exchanger geometry, evaluation of pressure drop, and dynamic simulation of the integrated system to assess temperature control performance under varying compressor load and cooling water conditions. The final solution should offer high thermal reliability, ease of integration, and improved operational resilience for the Shituru air compression system.

Key modules (Scientific and Engineering Knowledge):

Heat Transfer A (414) | Thermofluid Dynamics (344)

2 Mechatronic Topics at EE

Smart inverter control for voltage rise mitigation

Reference	: JBK1
Supervisor	: Dr Johan Beukes
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Operational small-scale embedded generators cause line voltage to rise, due to the direction of power flow. This rise currently limits the use of renewable energy on residential feeders to 25% of capacity before voltage limits are exceeded. If voltage can be controlled on these systems, this limit can be increased to higher levels. Inverters can control active and reactive power independently by controlling phase angle and voltage respectively. Smart inverters have the capability to change operating conditions based on communication signals. This project will start with investigating and quantifying the problem by acquiring information and data. An existing smart inverter will be installed and a communication link to an external computer will be established. A control strategy will be developed, tested in Matlab Simulink and then implemented using the assembled hardware.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Design (E) (314) | Electronics (315)

Microwave Sensor Loaded With Complementary Curved Ring Resonator for Material Permittivity Detection

Reference	: LB1
Supervisor	: Dr Leanne Bodenstein
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project focuses on the development of a microwave sensor utilizing a complementary curved ring resonator designed for the precise detection of material permittivity. By leveraging the unique electromagnetic properties of the curved ring resonator, the sensor aims to enhance sensitivity and accuracy in measuring dielectric constants across various materials. The sensor operates by analyzing changes in resonance frequency and bandwidth in response to the introduction of different substances, enabling real-time monitoring and characterization. Applications of this technology span multiple fields, including materials science, food quality assessment, and biomedical diagnostics, where understanding material properties is crucial. This innovative approach promises to advance non-destructive testing methods and improve the efficiency of material analysis processes. The student will have to design the sensor, manufacture and test the sensor and analyze the performance of the sensor.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electronics (245) | Electronics (315)

Evaluation of a Reverberation Chamber using Ray Tracing

Reference	: JG1
Supervisor	: Dr Jacki Gilmore
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

A reverberation chamber (RC) is a screened room (Faraday cage) used for electromagnetic investigations. Since an RC is effectively a high-Q cavity resonator, mode stirrers are used to lower the possibility of standing waves developing within the chamber. This project aims to employ ray tracing (as implemented in MATLAB, FEKO or CST) to evaluate the effectiveness of mode stirrers in scattering the EM waves in a reverberation chamber. Results can be verified using the Reverberation Chamber and RF measurement equipment in the microwave laboratory at E&E.

Key modules (Scientific and Engineering Knowledge):

Design of an Electrically Excited Synchronous Generator

Reference	: NG1
Supervisor	: Dr Nkosinathi Gule
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project focuses on the design and analysis of a rotor for an Electrically Excited Synchronous Generator (EESG). The work involves developing a rotor that efficiently incorporates field windings to generate a controllable excitation field, enabling improved torque production and operational flexibility. Key tasks include the analytical design and optimization of the rotor for minimal losses, and ensuring effective thermal management of the excitation windings. Finite Element Analysis (FEA) will be used to evaluate electromagnetic performance and flux distribution. The outcome of the project will be a robust, high-efficiency EESM rotor design suitable for modern electric generator applications.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Development of a High-Power Audio Amplifier with an Automated Thermal Management System

Reference	: TH1
Supervisor	: Ms Tessa Hall
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project involves the design and construction of an audio amplifier capable of delivering at least 75 WRMS into an 8 Ω load. The amplifier should feature integrated short-circuit protection for enhanced reliability and safety and a 3-channel graphic equaliser to allow precise control over bass, midrange, and treble frequencies. The thermal management system should monitor the thermal behaviour of the audio system and implement predictive control to cool the system when required. A real-time user interface for the system should be implemented.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Electronics (245) | Electronics (315)

CubeSat Attitude Determination Using Laser Retroreflector via Ground-Based Laser Ranging

Reference	: HWJ1
Supervisor	: Prof Willem Jordaan
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project focuses on developing a specialized Laser Retroreflector Array (LRA) to be mounted on a CubeSat, thereby overcoming the size and power constraints of traditional on-board attitude sensing by utilizing precise ground-based Satellite Laser Ranging (SLR) stations; the core innovation involves designing the LRA geometry—optimizing the placement and orientation of individual retroreflector elements—to ensure that the differential time-of-flight and distinct return signal patterns from the ground station can be processed by a computational algorithm to accurately infer the CubeSat's 3-axis attitude, thus providing a passive, highly accurate, and non-cooperative method for attitude determination critical for post-failure analysis or active debris removal missions. The project entails the development of the reflector/receiver system along with a laboratory experiment to demonstrate the working of the algorithms for attitude determination.

Key modules (Scientific and Engineering Knowledge):

Data Analytics (Eng) (344) | Design (E) (314) | Experimental Techniques (414)

Optical Tracking for High-Altitude Balloon Communication

Reference	: HWJ2
Supervisor	: Prof Willem Jordaan
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This engineering project aims to develop an optical system for actively controlling a ground-based tracking antenna to maintain robust communication with a high-altitude balloon (HAB) during the critical initial ascent phase of its mission, when rapid movement and complex atmospheric conditions pose significant challenges to traditional tracking methods. The project involves designing and implementing a vision-based tracker that utilizes real-time image processing of the balloon's visual signature to generate precise pointing commands for the communication antenna's pan-tilt mechanism, thereby ensuring continuous data telemetry and command up-link during the dynamic early stages of flight before stable, GPS-aided tracking can reliably take over. This system will enhance mission success rates by mitigating communication dropouts caused by rapid changes in balloon position and atmospheric obscuration, providing a vital bridge until the HAB reaches its operational altitude.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Data Analytics (Eng) (344) | Design (E) (314)

Design and evaluation of PM brushless exciter.

Reference	: MJK1
Supervisor	: Prof Maarten Kamper
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Brushless exciters play an important role in synchronous electrical machines where they provide power to the rotor windings. In this project we look at the design and evaluation of a 6-kW permanent magnet (PM) brushless exciter for a 200-kW, slow-rotating synchronous machine. The exciter must be analyzed by basic analysis and simulated with a diode rectifier as load using MOTORCAD finite element software. A small existing exciter must be tested in the laboratory and compared with MOTORCAD results.

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electrical Drive Systems (324)

Automatic Building/Terrain Profile Modelling for RF/EMF Assessments

Reference	: DLUDIC2
Supervisor	: Dr Danie Ludick
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Radio Frequency (RF) safety compliance assessments require simulating the antennas installed at a specific site (e.g., rooftop, tower, pole, etc.), and accounting for their transmitted power and electrical characteristics (such as antenna patterns and electrical tilt configurations). One of the most time-consuming steps in this process is creating a 3D model of the building, including the terrain profile and accurately placing the antennas.

The goal of this project is to streamline this step by automatically extracting building geometry (and nearby structures) from web-based data sources such as OpenStreetMap, the Google Earth SDK, and similar APIs. The project will involve developing a software utility that takes a GPS coordinate as input and then retrieves available building and terrain data for that location. Once the site's antennas are specified, the system will support performing an RF safety assessment using the imported spatial data.

Key modules (Scientific and Engineering Knowledge):

Data Analytics (Eng) (344) | Modelling (334)

Yield optimization of microwave structures

Reference	: PM1
Supervisor	: Prof Petrie Meyer
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Microwave devices are typically machined or etched, and suffer from dimensional errors which affect the electrical performance. In this project, a machine learning numerical model must be used in an optimization loop to optimize structures for yield and not only optimal nominal performance. The project is very mathematical, but will require only MATLAB programming.

Key modules (Scientific and Engineering Knowledge):

Data Analytics (Eng) (344) | Modelling (334)

Rally detection and automatic video capture for spikeball

Reference	: JCS1
Supervisor	: Dr JC Schoeman
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Spikeball (or roundnet) is an emerging sport that is mostly played at the amateur level. Much like volleyball, two teams of two players each have up to three touches to save, set and spike a round ball. The key differences are (a) that the ball is much smaller, (b) that the ball is bounced off of the net instead of hit over it, and (c) that there are no boundary lines or team-specific sides after the serve.

Video recordings of sport (especially at professional level) is very common, where some benefits include live broadcasting, making refereeing decisions, post-match reviewing for coaching staff, and simply making entertaining videos, to name a few. However, most of these solutions are very expensive, both due to the required hardware as well as the need to store large amounts of video data (and manually filter through it afterwards).

The aim of this project is to develop a low-cost system that can automatically detect spikeball rallies and record them when prompted by the user. More specifically, a smartphone app and/or Raspberry Pi setup needs to be developed that constantly stores (and overwrites) the last rally in a video buffer, and only saves it to permanent memory when the user indicates this desire, either by pressing a button (on the phone or a watch), giving a voice command or indicating otherwise using for example a hand signal. This is quite a challenge, since rallies are not always the same length and the configuration of the players can vary a lot, potentially even blocking certain camera angles. The solution will rely quite heavily on image processing and app/software development, as well as other peripherals as necessary.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (214) | Data Analytics (Eng) (344) | Design (E) (314)

Develop an automated waveguide VSWR tuner

Reference	: WES1
Supervisor	: Dr Werner Steyn
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Waveguides are used as high frequency transmission lines in high power applications. In these systems maximum power transfer to a complex load is crucial. The Voltage Standing Wave Ratio (VSWR), which is measured with a Vector Network Analyzer (VNA) in the lab, is an indication of the magnitude of the signal that is reflected from the load. The student will design a system that reads the VSWR from the VNA and automatically adjusts tunable elements in the waveguide to obtain minimum VSWR and maximum power transfer to a load.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Design (E) (314) | Electrical Drive Systems (324)

Test Rig for Developing Thermal Models of High Voltage Overhead Power Lines or Medium Voltage Cables

Reference	: AGS1
Supervisor	: Dr Andrew Swanson
Can be completed in first semester	: Yes
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Development of a test rig is for testing of high voltage conductors and/or medium voltage cables. The design will involve development of a high current power supply for the test rig using a phase angle controller and a PLC, and should allow for temperature cycling. Acquisition and logging of the current and temperature is required. The development of thermal models of the conductors and/or cables can be developed based on the results.

Key modules (Scientific and Engineering Knowledge):

Control Systems (354) | Heat Transfer A (414)

Partial Discharge Data Acquisition System

Reference	: AGS2
Supervisor	: Dr Andrew Swanson
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Partial discharge occurs in ageing high voltage systems. Partial discharge is a very small current with a short duration (ns-us) that needs to be measured with an appropriately sensitive tool. The student is to design a data acquisition system for partial discharge that accounts for the measurement, amplification, noise and filtering, and corresponding phase angle of the signal. The system should display the partial discharge patterns on an appropriate platform such as Matlab. (Note that a significant portion of the project does not require the use of high voltage equipment as it involves small signals that can be recreated in an electronics lab - but the equipment is available when required).

Key modules (Scientific and Engineering Knowledge):

Design (E) (314) | Electronics (315) | Experimental Techniques (414)

Thermal Analysis of an Axial-Flux Permanent-Magnet Machine: Evaluation of Advanced Cooling Techniques for Enhanced Power Density

Reference	: RJW1
Supervisor	: Prof. Rong-Jie Wang
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Axial-flux permanent-magnet (PM) machines are increasingly adopted in electric vehicles, drones, and other compact drives because of their high torque and power density. However, their flat, compact geometry presents significant challenges for heat removal. Conventional air or water-jacket cooling often limits continuous performance, motivating the development of more effective thermal management solutions. Recent studies have investigated advanced techniques such as direct oil cooling, dielectric-fluid immersion, and heat-pipe-assisted cooling to enhance heat extraction from windings and magnets while maintaining lightweight construction.

The aim of this project is to model and compare the thermal behaviors of a YASA-type axial-flux PM machine under both conventional and advanced cooling methods, in order to identify the configuration that achieves the highest sustainable power density.

Key modules (Scientific and Engineering Knowledge):

Electrical Drive Systems (324) | Thermofluid Dynamics (344)

Detection of Energy Inefficiencies in Mobile Cellular Network Base Stations

Reference	: RW1
Supervisor	: Prof Riaan Wolhuter
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

This project aims to develop a robust algorithm capable of detecting both sudden and gradual increases in energy consumption at mobile network base stations. Real-world energy meter data is available for approximately 5,000 sites, recorded at 30-minute intervals over a full year. Due to differences in site equipment and configurations, each location has a unique consumption profile. The algorithm must adapt to these variations and identify abnormal consumption trends by analysing historical data patterns.

The project will involve data cleaning and pre-processing, time series analysis, and the application of machine learning techniques. The final solution will be applied at scale and validated on real-world data. Sites flagged by the algorithm will be further investigated by the mobile network operator for potential energy inefficiencies.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Data Analytics (Eng) (344) | Design (E) (314)

Detection of leaf wetness in cultivated crops

Reference	: RW3
Supervisor	: Prof Riaan Wolhuter
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

The level of leaf wetness in agriculture is important in terms of plant health and potential disease. From references it appears that IR absorbtion by the plant is a good indication of leaf wetness. The proposal is to design and build a sensor device that would utilise IR spectrum sensing and level measurement, to evaluate the degree of leaf wetness, or the amount of water directly on leaves. The student will have to investigate whether an active IR source should be used, or natural IR radiation would be sufficient. Active testing and evaluation of the completed device will be required.

Key modules (Scientific and Engineering Knowledge):

Computer Systems (245) | Design (E) (314) | Electronics (315)